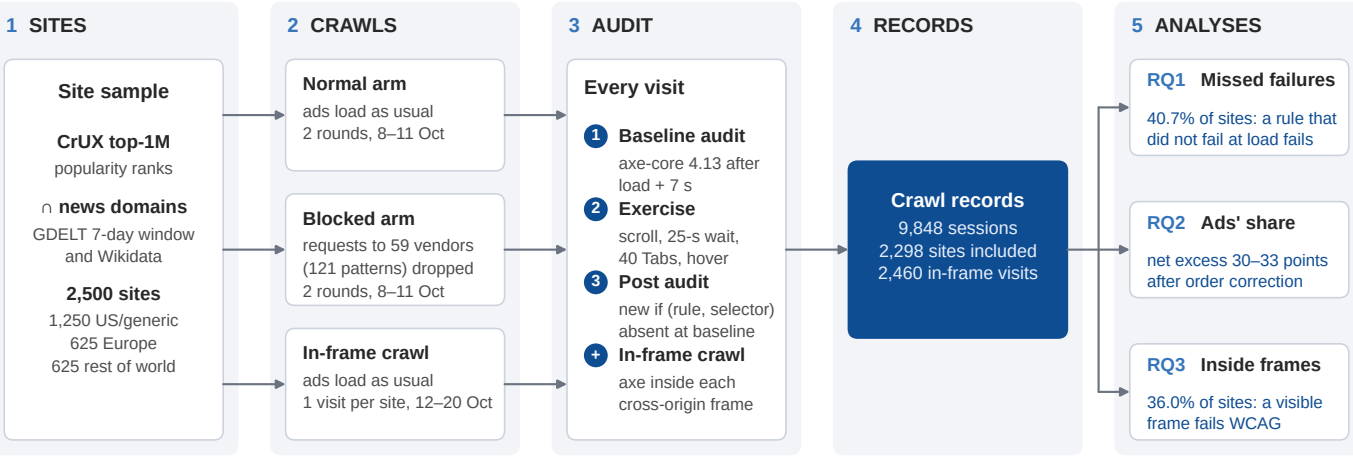
1 SITES
Site sample
CrUX top-1M
popularity ranks
∩ news domains
GDELT 7-day window
and Wikidata
2,500 sites
1,250 US/generic
625 Europe
625 rest of world
2 CRAWLS
Normal arm
ads load as usual
2 rounds, 8–11 Oct
Blocked arm
requests to 59 vendors
(121 patterns) dropped
2 rounds, 8–11 Oct
In-frame crawl
ads load as usual
1 visit per site, 12–20 Oct
3 AUDIT
Every visit
1 Baseline audit
axe-core 4.13 after
load + 7 s
2 Exercise
scroll, 25-s wait,
40 Tabs, hover
3 Post audit
new if (rule, selector)
absent at baseline
+ In-frame crawl
axe inside each
cross-origin frame
4 RECORDS
Crawl records
9,848 sessions
2,298 sites included
2,460 in-frame visits
5 ANALYSES
RQ1 Missed failures
40.7% of sites: a rule that
did not fail at load fails
RQ2 Ads' share
net excess 30–33 points
after order correction
RQ3 Inside frames
36.0% of sites: a visible
frame fails WCAG